Assignment 1 - Software Requirement Specification (UML Use Case Diagram, User Stories, and Functional Requirements)

Course code and name: ISYS3416/3432 - Software Engineering Fundamentals for IT

Length: not more than 20 pages

Type: Written Report

Feedback mode: Written Feedback

Late work:

If you require an Extension of Submittable Work (assignments, reports project work, etc.) for 5 calendar days or less (from the original due date) and have valid reasons, you must complete and lodge an Application for an Extension of Submittable Work (5 Calendar Days or less) form and lodge it with the Senior Educator/ Program Manager. The application must be lodged no later than one working day before the official due date. You will be notified within no more than 2 working days of the date of lodgment as to whether the extension has been granted. If you seek an Extension of Submittable Work for more than 5 calendar days (from the original due date), you must lodge an Application for Special Consideration form under the provisions of the Special Consideration Policy, preferably before, but no later than 2 working days after the official due date. A submittable work (assignments, projects, lab reports, etc.) that is submitted late without approval of an extension will not be accepted or marked.

Penalties for Late Submission: A penalty of 10% per day will apply to late assignments up to a maximum of five days (weekends count for two days, not one). Note that for assignments up to five days late due to illness, a medical certificate may be presented to your teacher, and a late penalty will not apply.

Learning Objectives Assessed

- CLO2: Evaluate requirements for a software system
- CLO3: Apply the process of analysis and design using the object-oriented approach
- CLO4: Work effectively in a team to analyse the requirements of a complex software system and solve problems by creating appropriate designs that satisfy these requirements.
- CLO5: Communicate effectively with others, especially regarding the progress of the system development and the content of the design, by means of reports and presentations. Use appropriate design, version control, and collaboration tools to work effectively as a team.
- CLO 6: Recognize current trends in the area of software engineering.

Ready for Life and Work

- The following skills will be developed upon completing this course such as:
  - **Modeling and Visualization Skills** – Translating abstract requirements into visual or textual models that stakeholders can easily interpret using case diagrams and enhance understanding of the system scope and actor interactions.

- **Communication and Collaboration Skills** include writing user stories in clear, concise, and user-centered language, as well as producing understandable requirement documents to bridge the gap between technical teams and non-technical stakeholders.
- **Analytical and Critical Thinking Skills** are used to distinguish between essential functionality (functional requirements) and system qualities (non-functional requirements).
- **Problem-Solving and Design Thinking Skills** are used to explore multiple ways to meet a user's need. Anticipating future scenarios and system scalability through non-functional requirements like performance, security, or usability.
- **Teamwork:** Work as an effective and productive team member in a range of professional and social situations, in particular to; work effectively in different roles, to form, manage, and successfully produce outcomes from teams, whose members may have diverse cultural backgrounds and life circumstances and differing levels of technical expertise.

Assessment Details

You need to form consultant teams (**composed of 4 members)** and produce the initial report for the assigned project.

Project Summary, Rationale, and Common Aspects

Software Engineering Fundamentals (SEF) is currently being taken by students with diverse backgrounds and programming experience. SEF is a core course in both programs (SE, IT). To accommodate such diversity, students will be assigned at random from a given project list, which is designed to allow for an agile or mixed approach for the first milestone and a more rigorous design for the second milestone using UML. We worked hard to come up with interesting and diverse projects to match your preferences. When working on the project, each team member should take into account the group's strengths, background, and interests. **The project title will be drawn by the lecturer during the tutorial session and assigned to your team.**

Assessment 1 (Milestone 1) is to understand the user requirements. These requirements describe what the user expects the system to do and the intended outcome. There are three perspectives to determine the user requirements:

- Functional requirements determine what the system should do. While the non-functional requirements describe how the system behaves and performs.
- Use Case diagrams express and outline the user's goals and expectations. Illustrate the activity flow of how the user interacts with the system and how the system responds to the user's action. To draw the diagrams, you can use any tools such as StarUML, UMLet, Lucid Chart, Miro, or Draw.io.
- User Stories will describe the feature or functionality from the user's perspective by focusing on the "who," "what," and "why".

In addition, the timeframe and estimations for completing the project should be outlined utilizing a software engineering methodology (e.g. Scrum, Waterfall, Hybrid).

To achieve this, each group will need to work through the tasks consistently and will also need to form subgroups to allow all tasks to be completed in parallel. One or two persons should be responsible for proofreading the document before submission to ensure consistency in language and format. Remember to check that numbering is sequential when used and that all headings are formatted similarly at the same level. Headers and footers should also be the same throughout.

You will write a document of the software requirement specification (SRS) for your assigned topic. Please produce a professional report. You will be graded on the presentation of the report. You can find the SRS template below to guide you in writing the report.

Note: As the Login and Logout functions will be frequent examples in the lectures and tutorials, team members are discouraged from using them as one of their identified use cases.

The list of project titles is as follows:

1. **Personalized Clothing Store**

The goal of this project is to design personalized clothing that allows customers to choose clothing items with custom text, images, or patches. Customers can browse the catalog of available clothing items and choose which ones they want to customize. They can use interactive tools to enter text, upload images, select patch placements, and view the finished product before adding it to their cart. Using a live camera, the user can see in real time whether the customized clothing fits their body structure. Before making any purchases, the customer must create an account on the platform.

2. **Cultural Heritage Information System using AR**

The goal of this project is to create a mobile application that uses Augmented Reality (AR) technology to enhance the cultural and historical exploration of specific churches in Vietnam. Using the mobile camera, the application will recognize architectural features within the selected churches. When viewed using a mobile device, the application will provide real-time contextual information about specific church features. The application can allow the user to save and share the captured image and information. Additionally, this application can serve as a digital tour guide and enhance the visitor experience.

3. **At-Risk Students Prediction**

The goal of this project is to provide early detection of at-risk students and timely intervention to improve their retention. Users can upload student records such as student information, courses, course registrations, assessments, and LMS activities like assignments and quizzes using the Power BI system. The dashboard will use predictive analytics to monitor student performance and engagement, identify students who are struggling academically or behaviorally, and alert lecturers to potential disengagement before it becomes a crisis, allowing for timely intervention. Based on the data, the system will create a dashboard that displays the results in a user-friendly, interactive format, allowing users to visualize trends, drill down into specific student or class data, and sort information to identify common issues.

4. **Hospital Emergency Response**

The goal of this project is to provide a mobile and web app for staff to instantly view and enter information as required. This app allows hospital administrators to add and manage system data required for the solution to work. The dashboard allows employees to view and analyze key data and metrics that will assist them in making more informed decisions. The system will use Power BI to display real-time data visualizations for critical metrics such as available beds, staffing levels, vital equipment, and discharge information. The system can customize the reports and dashboard to meet the user's specific requirements and integrate them with existing hospital applications and portals.

5. **Business Analytics Application**

The goal of this project is to create mobile-enabled analytics platforms that combine mobile app analytics, business intelligence (BI) tools, and market intelligence systems with real-time dashboard and reporting capabilities. The system should include user authentication and authorization, as well as a customizable dashboard with charts displaying KPIs, trend analysis, and drill-down features. The system can also generate reports with scheduling and sharing options. A real-time notification for anomalies or performance updates. Users can collaborate by commenting, annotating, or sharing dashboards within user groups.

Submission Instruction

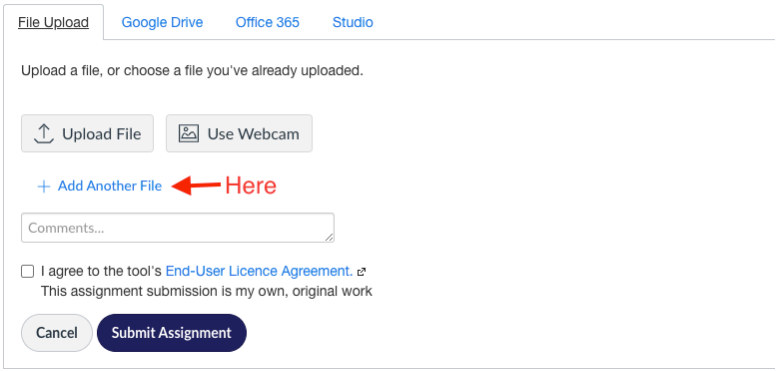
What to submit?

For your initial report, you will write a System Requirements Specification (SRS) document and a high-level UML-based system overview. Use the template provided to document your vision for the project. Download the document template here **A1 SRS Template**.

How to Submit?

1. The assignment must be submitted in Canvas with the naming convention:
   - For **SGS**: ISYS3416_A1_[GroupNumber]
2. The original UML diagram or a clear image file must be added as an additional file in Canvas (DO NOT zip all the additional files; submit it as an additional file). See the instructions below.

Other files are submitted separately by selecting "**add another file**"



Support Resources

This assessment requires that you meet RMIT's expectations for academic integrity. More information and advice on how to avoid plagiarism are available in the Getting Started module.

- Open **the academic integrity page**.

Additional library and learning resources are available to help with the assessment in this course

- Link to **Assignment Support**.

Criteria	Ratings					Pts
Introduction and Overall Description	**5 to >3.96 pts Excellent** Project purpose and scope are clearly and comprehensively defined. Overall description is complete and coherent, covering product perspective, functions, user classes, characteristics, and constraints with strong alignment and clarity.	**3.96 to >3.46 pts Good** Project purpose and scope are clear with minor gaps. Overall description addresses all required elements but lacks depth or precision in some areas.	**3.46 to >2.96 pts Satisfactory** Project purpose and scope are stated but lack detail or clarity. Overall description is present but some elements are vague or weakly connected.	**2.96 to >2.46 pts Fair** Project purpose and scope are unclear or incomplete. Overall description is limited, poorly structured, or missing key elements.	**2.46 to >0 pts Poor** Project purpose, scope, and overall description are missing, unclear, or inconsistent, with little meaningful information provided.	5 pts
Clarity and Completeness of User Stories	**10 to >7.93 pts Excellent** User stories are clear, complete, and well-structured. Each story includes a proper ID, a well-written statement (role, goal, reason), clearly articulated business value, and fully defined acceptance criteria that are specific, testable, and measurable. User stories are consistent, unambiguous, and ready for implementation. The user stories are fully aligned with the project objectives.	**7.93 to >6.93 pts Good** User stories are mostly clear and complete. IDs, statements, business value, and acceptance criteria are provided, though some elements may lack detail or precision. Acceptance criteria are generally testable but may require minor clarification.	**6.93 to >5.93 pts Satisfactory** User stories are present but lack clarity or completeness. Some elements (e.g., business value or acceptance criteria) are vague, incomplete, or weakly defined. Acceptance criteria are partially testable or inconsistently written.	**5.93 to >4.93 pts Fair** User stories are poorly written or incomplete, with vague statements or weak acceptance criteria.	**4.93 to >0 pts Poor** User stories are unclear, incomplete, or poorly structured. Key elements such as proper statements, business value, or acceptance criteria are missing or not testable. User stories are ambiguous and not ready for implementation.	10 pts

Criteria	Ratings					Pts
Detail and Accuracy of Use Cases	**10 to >7.93 pts Excellent** Use cases are clear, complete, and well-structured. Each use case includes a proper ID, descriptive title, clearly identified actors, well-defined preconditions, a logical and detailed main flow, relevant and clearly described alternative flows, and precise postconditions. The interaction is unambiguous, consistent, and fully supports system requirements.	**7.93 to >6.93 pts Good** Use cases are mostly complete and understandable. All key elements are included, but some parts (e.g., alternative flows or postconditions) may lack detail or clarity. Main flow is logical, and interactions are generally consistent.	**6.93 to >5.93 pts Satisfactory** Use cases are present but lack clarity or completeness. Some required elements are vague, missing, or poorly detailed. Main flow may be incomplete or inconsistently written, and alternative flows are weak or unclear.	**5.93 to >5.0 pts Fair** Use cases are poorly structured, missing multiple elements, or difficult to follow.	**5 to >0 pts Poor** Use cases are unclear, incomplete, or poorly structured. Several required elements are missing or incorrect. Flows are confusing, illogical, or do not adequately describe system–actor interactions.	10 pts
Quality and Specificity of Functional Requirements	**10 to >7.93 pts Excellent** Functional requirements are clearly, precisely, and formally stated using consistent and unambiguous language (e.g., “The system shall…”). All required functions are complete, logically organized, uniquely identifiable, and fully traceable to system objectives. Requirements	**7.93 to >6.93 pts Good** Functional requirements are clearly stated and mostly formal. Most required functions are complete and testable, though minor ambiguity, redundancy, or organizational issues may exist. Requirements are generally traceable to system objectives.	**6.93 to >5.93 pts Satisfactory** Functional requirements are present but lack full clarity or formal structure. Some requirements are vague, incomplete, poorly organized, or not easily testable. Traceability to system objectives is limited or unclear.	**5.93 to >4.93 pts Fair** Functional requirements are weakly stated, informal, or inconsistently written, reducing clarity and testability.	**4.93 to >0 pts Poor** Functional requirements are unclear, informal, incomplete, or missing. Statements are ambiguous, not testable, or inconsistent, and do not adequately describe system behavior or support project objectives.	10 pts

Criteria	Ratings					Pts
	are testable, verifiable, and free from redundancy or ambiguity.					
Non-Functional Requirements (NFRs)	**4 to >3.16 pts Excellent** Non-functional requirements are clearly and formally defined for all relevant quality attributes (e.g., performance, security, usability, availability). Requirements are specific, measurable, testable, and realistic (e.g., response time thresholds, uptime percentages, security controls). They are well organized, non-conflicting, and clearly aligned with system goals and user needs.	**3.16 to >2.76 pts Good** Most non-functional requirements are clearly stated and cover key quality attributes. Requirements are generally measurable and testable, though some lack precision or complete metrics. Alignment with system goals is mostly clear.	**2.76 to >2.36 pts Satisfactory** Non-functional requirements are present but incomplete or vaguely stated. Several quality attributes lack clear metrics or testability. Requirements may be generic, loosely defined, or inconsistently organized.	**2.36 to >1.96 pts Fair** Non-functional requirements are weak, inconsistently defined, or largely descriptive without clear metrics.	**1.96 to >0 pts Poor** Non-functional requirements are missing, unclear, or purely descriptive. Quality attributes are not measurable or testable, and requirements do not clearly support system goals or constraints.	4 pts
Traceability and Consistency	**3 to >2.37 pts Excellent** Excellent linkage among user stories, use cases, functional requirements, and test cases. The traceability matrix is complete, accurate, and clearly documented, demonstrating	**2.37 to >2.07 pts Good** Strong linkage among user stories, use cases, functional requirements, and test cases. The traceability matrix is mostly complete and accurate, with minor gaps or inconsistencies. Bidirectional	**2.07 to >1.77 pts Satisfactory** Adequate linkage among most artifacts, but some connections are missing or unclear. The traceability matrix is present but partially complete or inconsistently maintained. Bidirectional	**1.77 to >1.47 pts Fair** Weak or inconsistent linkage among requirements and test artifacts. The traceability matrix is incomplete, poorly organized, or contains several errors. Traceability is mostly one-	**1.47 to >0 pts Poor** Little to no linkage among user stories, use cases, functional requirements, and test cases. The traceability matrix is missing or inaccurate, with no clear evidence of	3 pts

Criteria	Ratings					Pts
	full bidirectional traceability (forward and backward coverage) with no gaps or inconsistencies.	traceability is largely evident.	coverage is limited.	directional or unclear.	traceability or consistency.	
Activity Diagram	**8 to >6.31 pts Excellent** Activity diagram is complete and accurately represents all key workflows derived from the use case diagram. Correct UML notation is consistently applied (e.g., start/end nodes, actions, decisions, forks/joins, swimlanes). The workflow clearly reflects system behavior and aligns fully with the use cases. The description is concise, clear, and effectively explains the process flow and decision points.	**6.31 to >5.51 pts Good** Activity diagram represents most workflows based on the use case diagram with generally correct UML notation. Minor omissions or small notation issues may exist. The description explains the process clearly but may lack some detail or clarity.	**5.51 to >4.71 pts Satisfactory** Activity diagram covers main activities but may omit some steps or actors from the use case diagram. Some UML notation errors or simplifications are present. The description provides a basic explanation but lacks clarity or completeness.	**4.71 to >3.92 pts Fair** Activity diagram is partially complete or weakly aligned with the use case diagram. Significant UML notation errors or missing workflow elements are present. The description is vague, unclear, or insufficient.	**3.92 to >0 pts Poor** Activity diagram is missing or fundamentally incorrect. Major UML notation errors or no clear connection to the use case diagram. Description is missing or does not explain the process.	8 pts
						Total Points: 50